MILITARY SPECIFICATION

DU-DL DRIVER (CIRCUIT BOARD)
8623302

1. SCOPE

1.1 This specification covers one type of DU-DL Driver Circuit Board, part of the Test Set, Computer Logic Unit AN/GSM-70.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel, General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Ordnance Materiel: Packaging of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of
MIL-C-46357	Computer, Gun Direction: M18; Manufacturing Processes and Inspection of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts

DRAWINGS

U.S. Army Munitions Command

F8623302	DU-DL Driver Assembly
F8623303	Schematic, Electrical

PACKAGING DATA SHEET

M14002	Data Sheet, Master Packaging

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(Copies of specifications, standards, drawings, and packaging data sheets, required by the suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Fabrication.- The DU-DL driver circuit board shall be manufactured in accordance with Drawing F8623302 and drawings pertaining thereto.

3.2 General specification.- The DU-DL driver circuit board shall meet the following requirements of MIL-F-13926 and MIL-C-46357.

3.2.1 MIL-F-13926

a. Order of precedence
b. Dimensions and tolerances
c. Inorganic protective surface finishes
d. Workmanship

3.2.2 MIL-C-46357

a. Etched circuit assembly
b. Marking
c. Handling

3.3 Environmental service conditions.

3.3.1 Vibration.- The circuit board shall be subjected to a simple harmonic motion having an amplitude of 0.015 inch (0.030 inch total excursion). The frequency shall be varied uniformly from 10 to 55 cycles per second (cps) and return to 10 cycles. This cycle shall be traversed in 1 minute $\pm$ 15 seconds. The motion shall be applied to the circuit board for a period of 10 minutes in each direction of the 3 mutually perpendicular axes (total of 30 minutes), relative to its normal operational position. Subsequent to vibration, the circuit board shall not be damaged or adversely affected and shall meet the requirements of 3.4.

3.3.2 Shock.- The circuit board shall withstand 3 shock impluses in the vertical plane relative to its normal mounting position. The acceleration of the shock impulses shall be 5 gravity units (G's). Pulse duration of each impulse shall be 11 $\pm$ 1 milliseconds. Subsequent to shock the item shall not have been damaged or adversely affected and shall meet the requirements of 3.4.

3.3.3 Temperatures.

3.3.3.1 Operating.- The circuit board shall meet the requirements of 3.5 while exposed to temperatures of -40°F and +135°F.

3.3.3.2 Storage.- The circuit board shall meet the requirements of 3.4 at a temperature of between 60°F and 90°F after having been exposed to each temperature of -80°F and +160°F.

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3.3.4 Humidity.

3.3.4.1 Operating.- The circuit board shall meet the requirements of 3.5 while exposed to an atmosphere with a relative humidity (RH) up to 97% at a temperature of 135°F.

3.3.4.2 Non-operating.- The circuit board shall meet the requirements of 3.4 after exposure to (and when thoroughly dried) an atmosphere with a RH of 100 percent including condensation, for a period of 4 hours.

3.3.5 Fungus resistance.- The circuit board shall show no evidence of damage and shall meet the requirements of 3.4, after being inoculated with the composite spore suspension specified in Specification MIL-F-13926 and incubated at an ambient temperature of between 77°F and 84°F with a RH of 97 ± 2%.

3.3.6 Salt spray.- The circuit board shall show no evidence of physical damage and meet the requirements of 3.4 immediately after exposure to a fine spray of a 5% salt solution concentration. The solution shall have a specific gravity of from 1.0268 to 1.0413 and a pH between 6.5 and 7.2 at a temperature between 93° and 97°F.

3.4 Electrical.

3.4.1 DU-DL Driver.- With appropriate voltages depicted on electrical schematic, Drawing F8623303, applied and the input gate, shown in Figure 1, connected to the base of the input transistor, and the load shown in Figure 2 connected to the collector of the output transistor, each driver circuit shall function as specified below.

3.4.1.1 True state condition.- With a minus 6 volts plus or minus 0.18 volts applied to the input gate, the following conditions shall prevail:

(a) The voltage at the collector of the input transistor with respect to the -1.5 volt level shall be between zero and a minus 0.2 volts.

(b) The voltage at the base of the output transistor with respect to power ground shall be between a plus 4.3 volts and plus 5.0 volts.

(c) The voltage at the collector of the output transistor with respect to power ground shall be between a minus 5.8 and a minus 7.0.

3.4.1.2 False state condition.- With a minus 0.15 plus or minus 0.15 volt applied to the input gate, the following conditions shall prevail:

(a) The voltage across the zener diode shall be between 5.5 volts and 6.5 volts.

(b) The voltage at the base of the output transistor with respect to power ground shall be between zero and minus 0.64 volt.

(c) The voltage at the collector of the output transistor with respect to power ground shall be between zero and minus 0.5 volt.

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3.4.1.3 Dynamic operation.- Application of the 10KC input signal shown in Figure 3 shall produce a wave output with the characteristics as shown in Figure 3 for each of the drivers.

3.5 Operating temperature extremes.- Each driver circuit shall be capable of performing in accordance to the requirements described in 3.4.1.3.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component inspection requirements of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 Inspection provisions.

4.2.1 Submission of product.- Unless otherwise specified herein, or by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" as specified in MIL-STD-105.

4.2.2 Examinations and tests.- Except where otherwise specified herein, components and sub-assemblies shall be inspected in accordance with the "Inspection provisions for components and sub-assemblies" and the Sampling Plan Tables contained in MIL-F-13926. Examinations and tests related to Section 3 herein shall be performed in accordance with Tables I, II and III and the test methods and procedures in 4.5. Examinations and tests for packaging, packing and marking shall be in accordance with MIL-P-14232 and Section 5 herein.

4.2.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.2.4 Functional tests.- The requirements and tests in Table I shall be inspected on a 100% basis. All examinations and tests shall be conducted at the standard ambient temperature (60° to 90°F).

TABLE I
(100 percent inspection)

Characteristic	Requirement	Test Procedures
101. True state condition	3.4.1.1	4.5.2.2
102. False state condition	3.4.1.2	4.5.2.3
103. Dynamic operation	3.4.1.3	4.5.2.4

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4.3 Control tests.

4.3.1 General.- One circuit board shall be selected at random by a Government representative as a control sample from each 100 produced. The control sample shall meet the requirements and tests in Table II.

TABLE II

CONTROL TEST	REQUIREMENT	TEST PROCEDURES
301. Fabrication	3.1	Applicable drawings - visual
302. General specifications	3.2	MIL-F-13926 and MIL-C-46357 - visual

4.3.2 Environmental.- Three units shall be selected at random by a Government representative as a control sample from each 50 produced or from each month's production, whichever occurs first. The control samples shall have met the requirements and tests in Table I and shall then meet the requirements and tests in Table III.

TABLE III

CONTROL TEST	REQUIREMENT	TEST PROCEDURES
303. Vibration	3.3.1	4.5.1.1
304. Shock	3.3.2	4.5.1.2
305. Humidity-Operating	3.3.4.1	4.5.1.3.2
306. Humidity-Non-operating	3.3.4.2	4.5.1.3.1
307. Salt spray	3.3.6	4.5.1.4
308. Operating temperature	3.3.3.1, 3.5	4.5.1.5, 4.5.3
309. Storage temperature	3.3.3.2	4.5.1.5

4.3.3 Control test failure.- Should any one item of a control test sampling fail to meet the specified test requirements, acceptance of the product will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved (See 4.2.3).

4.4 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply equipment in accordance with the applicable requirements of MIL-I-45607. The Government reserves the right to use the test equipment for its own independent inspections to the extent that such use will not unduly interfere with the contractor's delivery schedule.

4.4.1 Contractor design.- The contractor shall provide inspection equipment compatible with the "Test Methods and Procedures" specified in 4.5 and with the "Component Inspection" and "Test facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

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4.4.1.1 Test equipment.- In conjunction with 4.4.1, the following test equipment shall be utilized in the performance of the applicable tests as specified in 4.5.

Item Number	Description
1	Power Supply: minus 1.5 volts nominal current rating greater than or equal to 650 milliamperes.
2	Power Supply: plus 15 volts nominal current rating greater than or equal to 100 milliamperes.
3	Power Supply: minus 6 volts nominal current rating greater than or equal to 200 milliamperes.
4	Power Supply: minus 18 volts nominal current rating greater than or equal to 600 milliamperes.
5	Voltmeter, D.C.Digital: A range of at least .001 to 999 volts d.c., or three significant figures. Accuracy of at least plus or minus 0.2% of reading ± 1 count; input impedance of at least 11 megohms on all ranges.
6	Generator, square wave: Shall be capable of delivering a symmetrical square wave at a frequency of 10 kilocycles per second, and a voltage swinging between zero volts and minus 6 volts. The rise and fall times of the output wave shall be no greater than 0.4 microseconds. The output impedance shall be no greater than 100 ohms.
7	Oscilloscope: Dual Channel; rise time of scope plus measuring probe 50 millimicroseconds or less; sweep period 50 millimicroseconds per centimeter or greater; input resistance greater than 8 megohms; input capacitance less than 25 microfarads.

(Items 1 through 4 shall be variable at least through the region 20 percent above and below the nominal value. These items shall have a regulation not greater than plus or minus 2 percent; a peak-to-peak ripple less than 1 percent; and an output impedance less than 2 ohms from d.c. to 500kc.)

4.5 Test methods and procedures.

4.5.1 Environmental.

4.5.1.1 Vibration.- The vibration test shall be made on a device with means provided for measuring and controlling the direction of vibration and adjusting the amplitudes and frequencies to keep them within the specified limits of 3.3.1. The measuring instruments of the testing device shall be capable of measuring instruments of the testing device shall be capable of measuring the vibration

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amplitude within plus or minus 5 percent, and capable of measuring the frequency within plus or minus 2 percent. The test shall be performed in accordance with the requirements specified in 3.3.1 with the circuit board rigidly mounted in its normal operating position. Subsequent to the vibration test, the unit shall show no damage or defect resulting from the test when examined by visual and tactile means. The unit shall then be inspected as specified in 4.5.1.2 prior to determining compliance with requirements specified under 3.4.

4.5.1.2 Shock.- This test shall be conducted in accordance with Procedure II of the shock test defined in MIL-F-13926 except that the circuit board shall be shocked in the vertical plane only. The impact (shocks) and "G" value shall be in accordance with 3.3.2. A receiving device, representing the connector receptacle that receives the circuit board in the computer, shall be secured to the horizontal surface of the shock tester. Each end of the receiving device shall be clamped such that the female receiving slot will hold the unit in a vertical plane. The unit will then be inserted into the receiving device and shocked. Subsequent to imposing the number of impacts required, a visual examination shall be made for damages or defects to determine compliance with 3.3.2 and then the unit will be inspected as specified in 4.5.1.3 prior to determining compliance with 3.4 as required by 3.3.2.

4.5.1.3 Humidity.

4.5.1.3.1 Non-operating.- This test shall be conducted in accordance with the conditions in 3.3.4.2 and the humidity test defined in Procedure I of MIL-F-13926 except that the 4-1/2 hour period shall be reduced to 2 hours, the 6 hour period reduced to 2 hours, and the total test period shall not exceed 5.5 hours per cycle. The unit shall then be thoroughly dried and then subjected to the test in 4.5.1.3.2 prior to meeting the specified requirements of 3.4 as required by 3.3.4.2.

4.5.1.3.2 Operating.- This test shall be conducted in accordance with the conditions in 3.3.4.1 and the humidity test defined in Procedure III of MIL-F-13926 except that the humidity shall be maintained at 95 $\pm$ 5 percent for the 24 hour period, and the test shall cease at the end of that period (Method 106 of MIL-STD-202 shall be excluded). Throughout the test the unit shall be operated in accordance with 4.5.3, to determine compliance with 3.3.4.1.

4.5.1.4 Salt spray.- The circuit board shall be subjected to the conditions in 3.3.6 and the salt spray test defined in MIL-STD-202, Condition B. Immediately after exposure, the unit shall be inspected for any physical damage, then tested in accordance with 4.5.2 for compliance with 3.3.6.

4.5.1.5 Temperature (Operating and storage).- After being subjected to all environmental tests specified above, the circuit board shall be subjected to Test Procedure I specified in MIL-F-13926, except the temperatures specified in 3.3.3 shall apply. While at the operating temperatures of 3.3.3.1, the unit shall be tested in accordance with 4.5.3 to determine compliance with the requirements of 3.5 as required by 3.3.3.1. The temperature shall then be returned to standard temperature (60° to 90°F) and the unit shall be tested in accordance with 4.5.2 to 4.5.2.4 inclusive to determine compliance with 3.3.3.2.

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4.5.1.6 Fungus.- To determine compliance with 3.3.5, one of the first 5 DU-DL drivers produced shall be submitted for testing in accordance with the fungus test defined in MIL-F-13926. After approval, subsequent acceptance for fungus requirements may be by certification.

4.5.2 Performance, Electrical.

4.5.2.1 DU-DL biasing voltages: Compliance with 3.4 shall be determined by the following test:

(a) Connect the voltage supplies indicated in Column A of Table IV to the pins in Column B.

TABLE IV

A	B
-1.5 ± .05 volts (Item 1)	4A, 13B, 17B, 24B, 31B, 34B, 39B, 47B, 54B, 60B
-18 ± 0.5 volts (Item 4)	20B, 46B
+15 ± 0.5 volts (Item 2)	33B
-6 ± 0.18 volts (Item 3)	33A

(b) The dc return of each of the voltage supplies shall be connected to pins 30A, 31A, 35A, and 36A.

(c) The voltage supplies shall remain connected for all electrical tests.

4.5.2.2 True state test.

(1) Connect a minus 6 volts plus or minus 0.18 volts to the input terminal of the input gate shown in Figure 1.

(2) Connect an input gate shown in Figure 1 to the pins in Column A of Table V, one at a time, and connect the load shown in Figure 2 to the corresponding pin in Column D.

(3) Read the voltage at the corresponding pin in Column B of Table V with respect to minus 1.5 volt level. This voltage shall be between zero and a minus 0.2 volts.

(4) Read the voltage at the corresponding pin in Column C of Table V with respect to power ground. This voltage shall be between a plus 4.3 volts and a plus 5.0 volts.

(5) Read the voltage at the corresponding pin in Column D with respect to power ground. This voltage shall be between a minus 5.8 volts and a minus 7.0 volts.

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4.5.2.3 False state test.

(1) Connect the input terminal of the input gate shown in Figure 1 to power ground.

(2) Connect an input gate shown in Figure 1 to the pins in Column A, one at a time, and connect the load shown in Figure 2 to the corresponding pin in Column D.

(3) Read the voltage at the corresponding pin in Column B with respect to the corresponding pin of Column C of Table V. This voltage shall be between a minus 5.50 volts and a minus 6.50 volts.

(4) Read the voltage at the corresponding pin in Column C with respect to power ground. This voltage shall be between zero and a minus 0.64 volts.

(5) Read the voltage at the corresponding pin in Column D with respect to power ground. This voltage shall be between zero and a minus 0.5 volt.

(6) All twenty circuits shall be so tested.

TABLE V

Driver Circuit No.	A (Input)	B	C	D (output)
1	1B	2B	1A	2A
2	3B	4B	3A	6A
3	5B	6B	7B	7A
4	9B	10B	8B	9A
5	11B	12B	13A	14A
6	14B	15B	16B	16A
7	18B	19B	20A	21A
8	22B	23B	21B	22A
9	25B	26B	26A	28A
10	28B	29B	30B	29A
11	37B	38B	36B	37A
12	41B	42B	39A	38A
13	43B	44B	45B	42A
14	48B	49B	50B	45A
15	52B	53B	51B	46A
16	55B	56B	51A	50A
17	57B	58B	57A	52A
18	61B	62A	59B	58A
19	62B	63B	63A	59A
20	64B	65B	65A	64A

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4.5.2.4 Dynamic check.

4.5.2.4.1 Test signal.- Using the square wave generator (item 6), an approximately symmetrical square wave (each half period shall be 50 plus or minus 2 microseconds), varying from 0.2 plus or minus 0.2 volts to minus 6 plus or minus 0.5 volt with 10 to 90 percent rise and fall times each less than 0.4 microseconds, shall be produced.

4.5.2.4.2 Dynamic test - driver loaded.

(a) Ground one terminal of the square wave generator and connect the other terminal to the input terminal of the input gate shown in Figure 1.

(b) Connect the input gate of Figure 1 to the pins in column A of Table II one at a time, and connect the load shown in Figure 2 to the corresponding pin in column D.

(c) Using the oscilloscope (item 7), observe the waveform of the voltage from the corresponding pin in column D to power ground. This voltage waveform shall be a square wave swing between the levels of minus 0.25 plus or minus 0.25 volts and a minus 6 plus or minus 1.0 volts at a frequency of the input test signal.

(d) Using the oscilloscope (item 7) compare the waveform from the corresponding pin in column D to the voltage waveform at the input terminal of the input gate, each with respect to power ground. The delay time measured from the time the square wave applied to the input reaches minus 1.5 volts (going toward minus 6 volts) until the output reaches 90 percent of its total excursion (going towards minus 6 volts) shall be less than 5 microseconds. The delay time measured from the time the square wave applied to the input reaches minus 1.5 volts (going toward zero volts) until output reaches 90 percent of its total excursion (going toward zero volts) shall be less than 5 microseconds (see Figure 3).

4.5.3 Extreme temperature tests.- Compliance to the temperature requirements of 3.5 shall be determined by the following tests:

4.5.3.1 High temperature extreme test.- Connect the voltage supplies indicated in Column A of Table VI to the pins in the Column B. Repeat 4.5.2.4.

4.5.3.2 Low temperature extreme test.- Connect the voltage supplies indicated in Column A of Table VII to the pins in Column B. Repeat 4.5.2.4.

TABLE VI

A	B
-1.40 volts ± 2% (Item 1)	4A, 13B, 17B, 24B, 31B 34B, 39B, 47B, 54B, 60B
-16.7 volts ± 2% (Item 4)	20B, 46B
+14.0 volts ± 2% (Item 2)	33B
-5.58 volts ± 2% (Item 3)	33A

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TABLE VII

A	B
-1.61 volts ± 2% (Item 1)	4A, 13B, 17B, 24B, 31B 34B, 39B, 47B, 54B, 60B
-19.3 volts ± 2% (Item 4)	20B, 46B
+16.1 volts ± 2% (Item 2)	33B
-6.42 volts ± 2% (Item 3)	33A

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging and packing.- Preservation, packaging and packing shall be in accordance with Master Packaging Data Sheet M14002.

6. NOTES

6.1 Intended use.- The DU-DL Driver is a non-phase inverting power amplifier used in the Test Set, Computer Logic Unit AN/GSM 70() to drive logic circuits.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Selection of applicable level of preservation, packaging and packing.
(c) Applicable stock number.
(d) Applicable packaging data sheet.
(e) A block of serial numbers to identify each circuit board.

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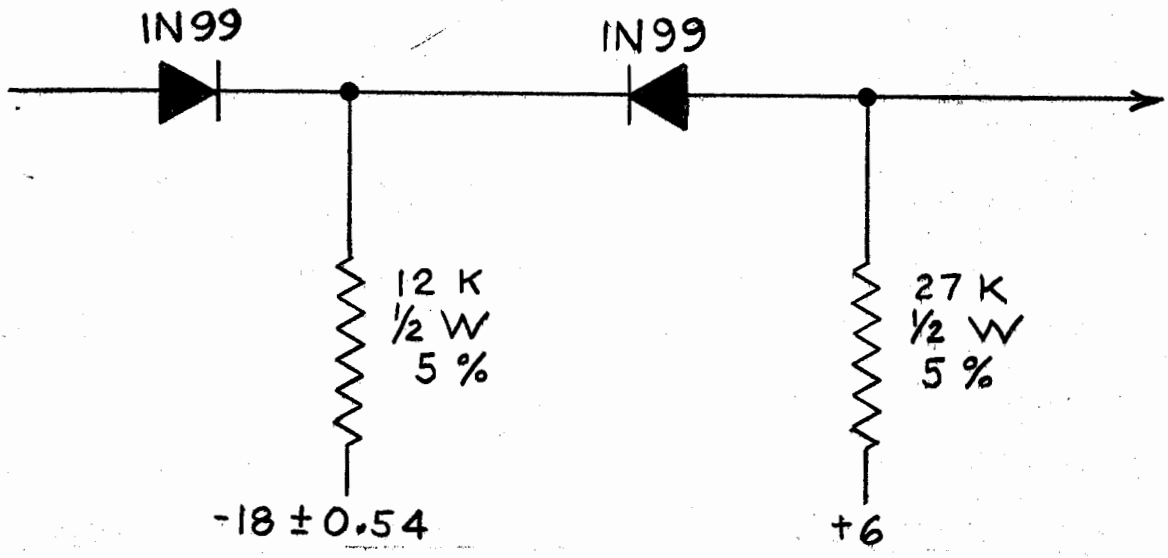


FIGURE 1

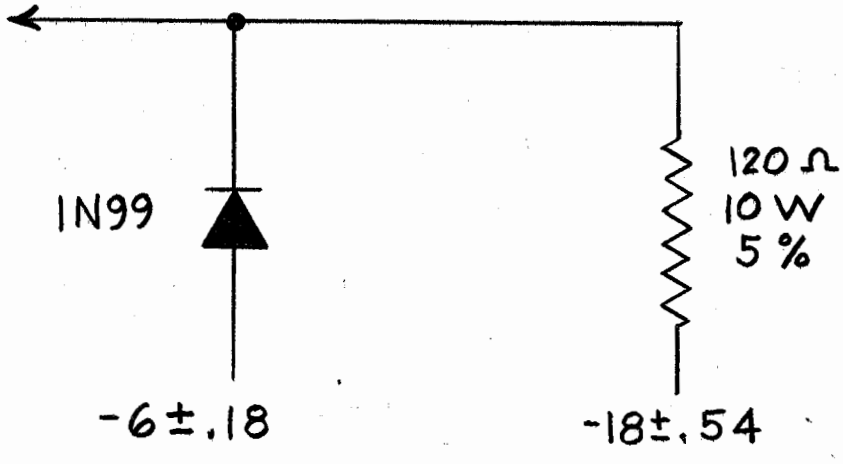
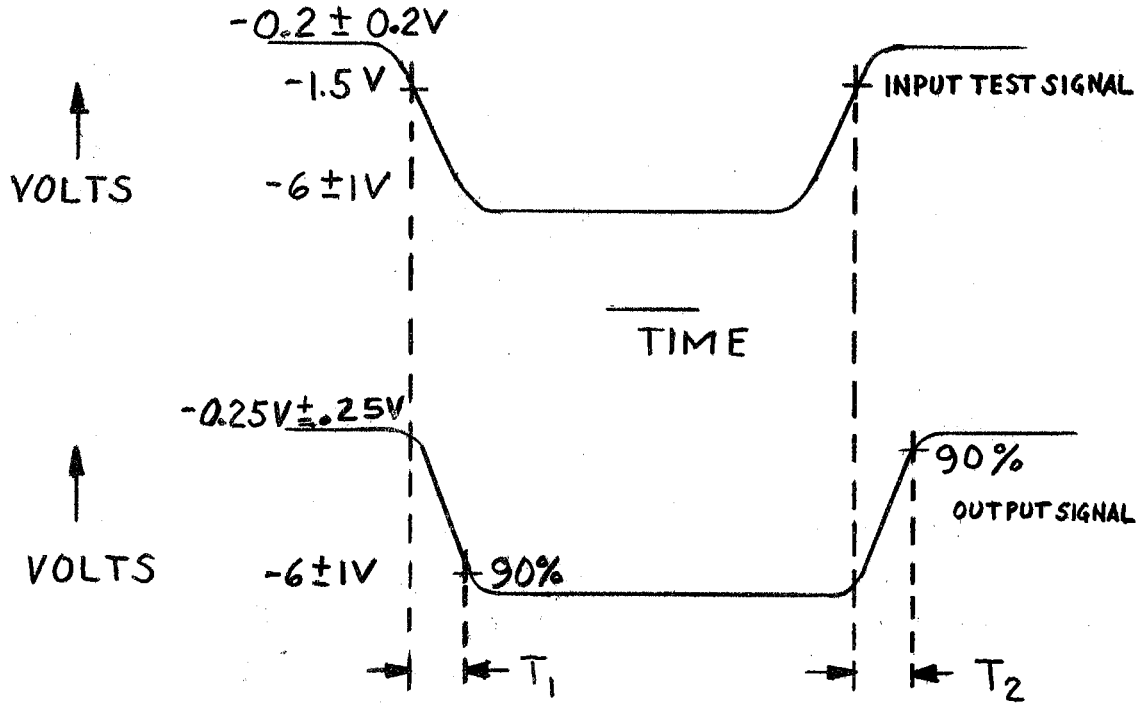


FIGURE 2

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TEMPERATURE CONDITIONS

ROOM AMBIENT

+135° F

-40° F

$T_1, T_2 \leq 5$ M SEC.

$T_1, T_2 \leq 25$ M SEC.

$T_1, T_2 \leq 10$ M SEC.

FIGURE 3

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